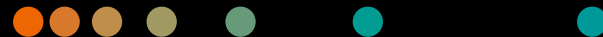
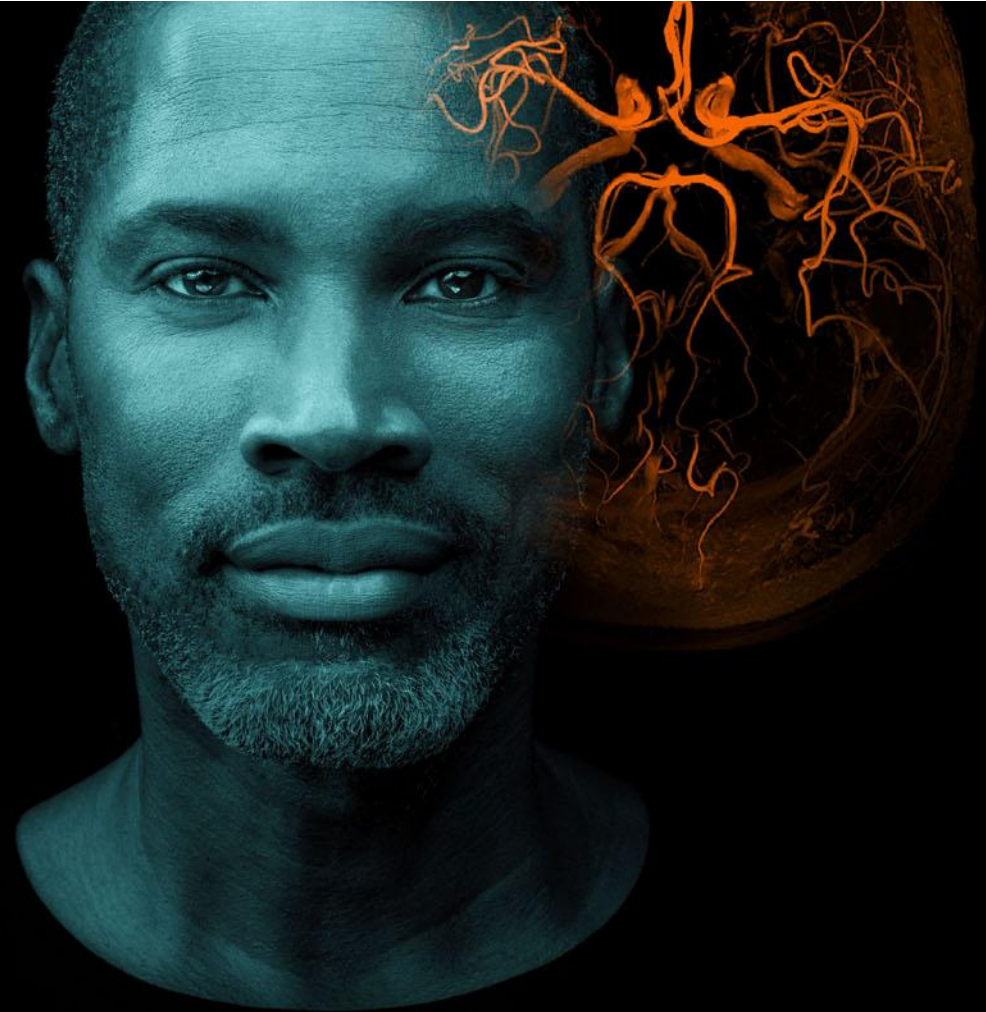


2026 Intern Showcase

Kyuwon Weon



About Me

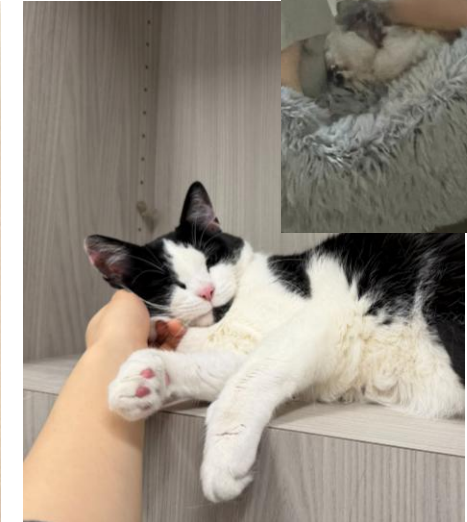
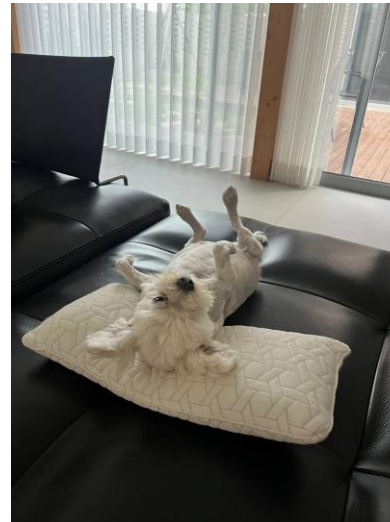
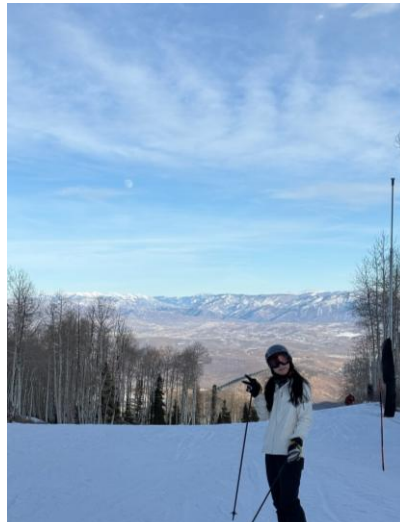
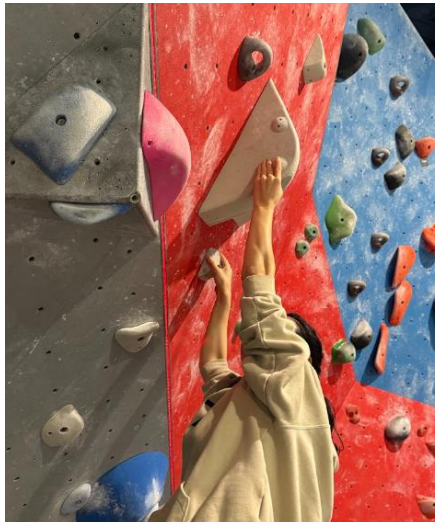
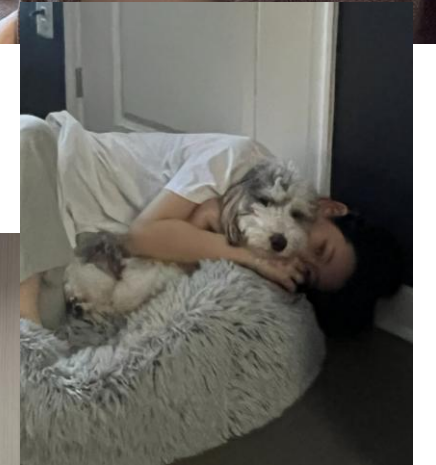
Kyuwon Weon

Manufacturing Intern @ Siemens Healthineers

M.S in Robotics @Northwestern (Dec 2026)

B.S in Mechanical Engineering

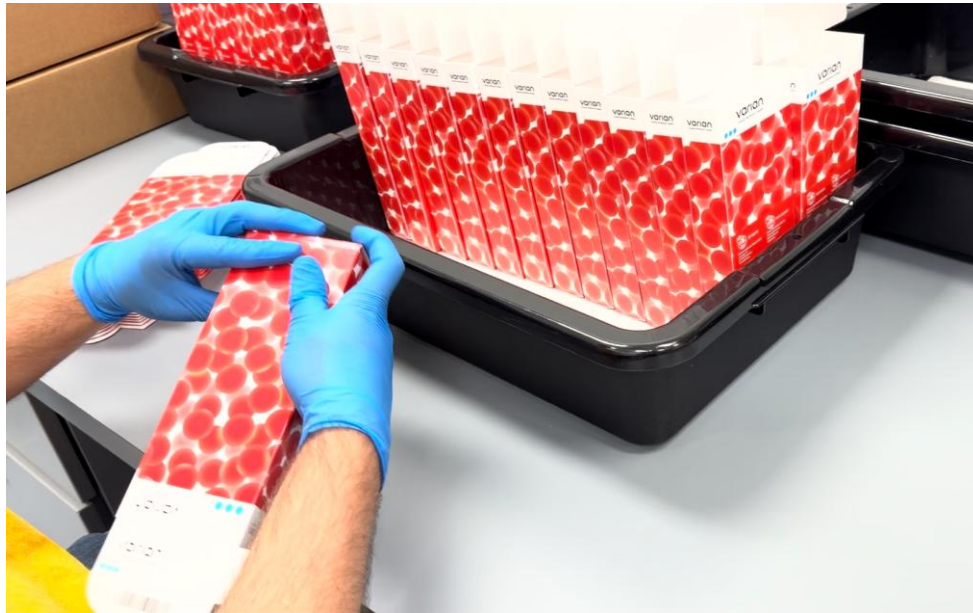
+ Biomedical Engineering @CMU '22



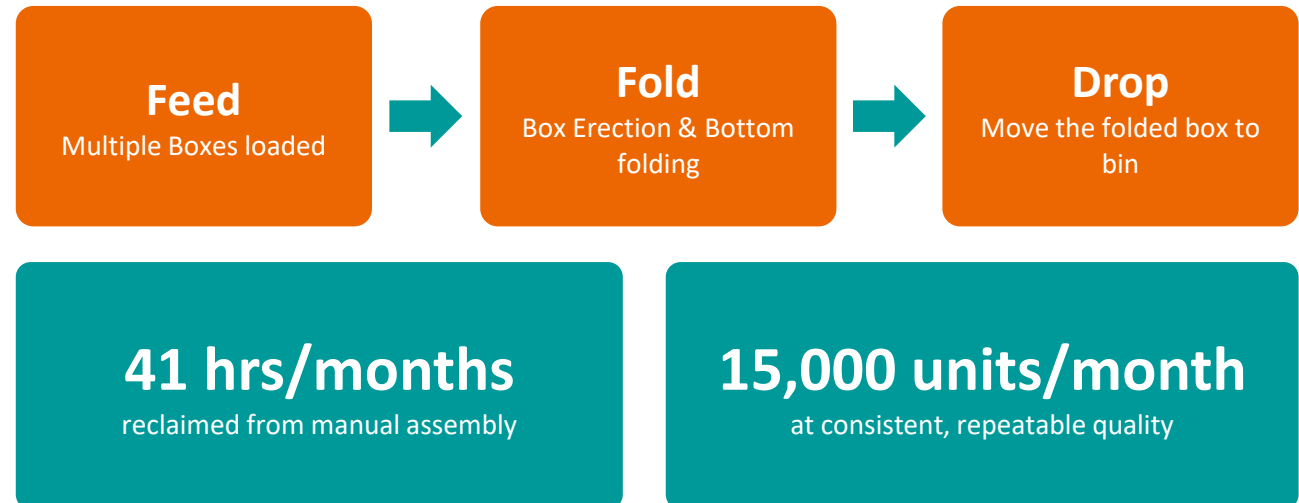
Packaging Automation

Problem Statement

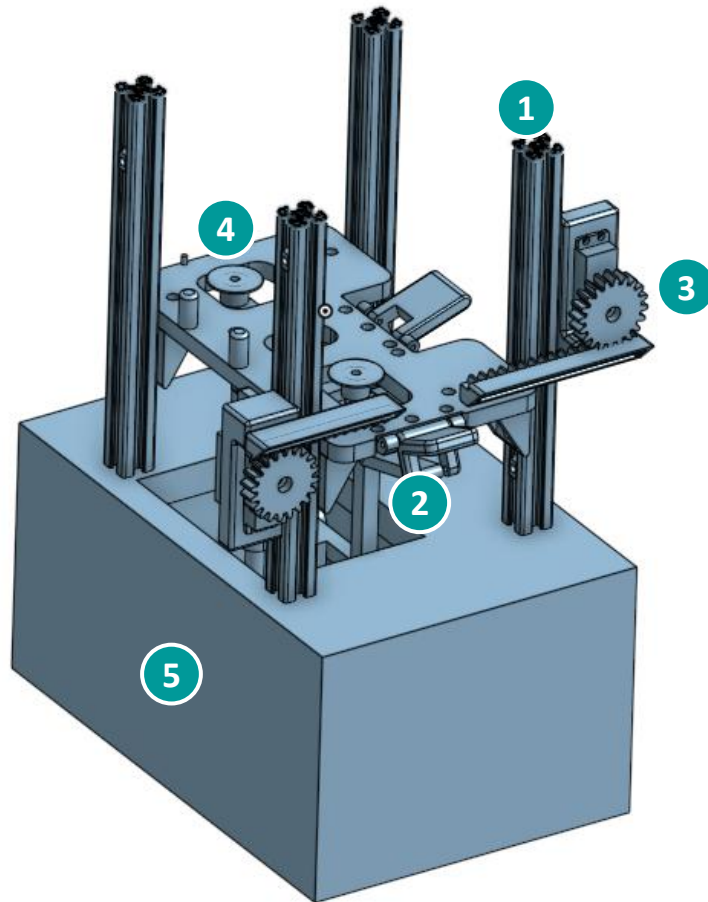
Folding boxes by hand takes 10 sec/box across 15,000 units per month, tying up **41 hrs/mo** in repetitive labor. Automating this process would free the time for higher-value work and reduce physical strain due to repetitive motion.



Solution Overview



Mechanical Design – Folding



Key Components

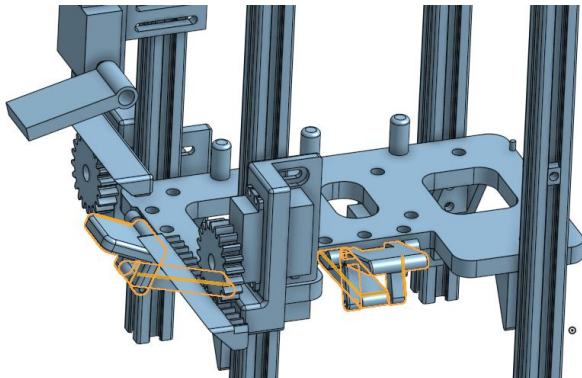
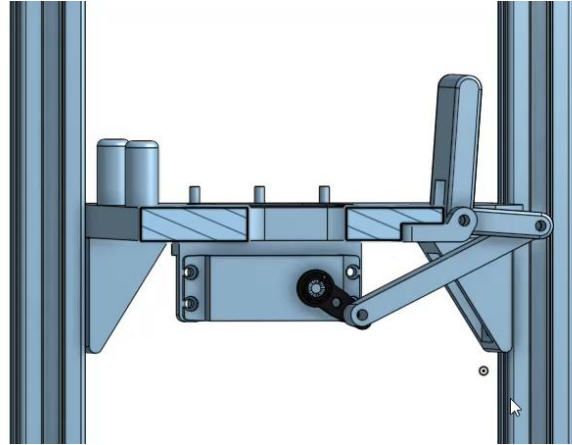
- 1 Frame — 2020 T-slot rails on a 3D-printed base with alignment features
- 2 Servo motor — drives the folding actuation
- 3 Rack & pinion — converts servo rotation into linear motion
- 4 Vacuum pump — holds the box securely in position
- 5 Base enclosure — houses electronics and mounting

BOM

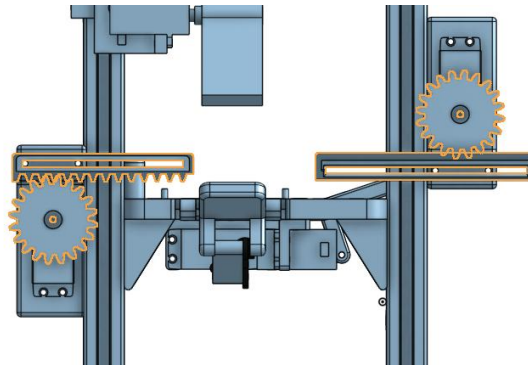
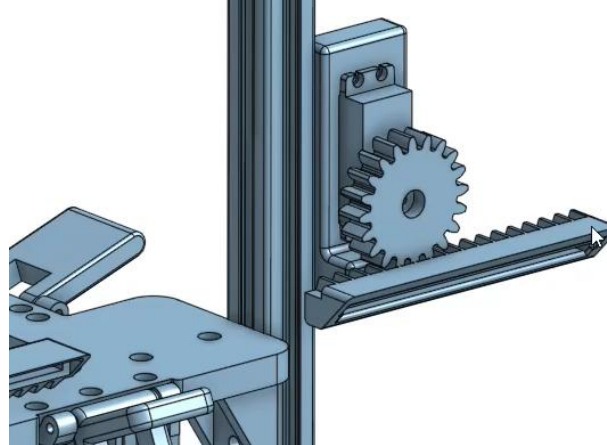
[Package automation bom.xlsx](#) (Screenshot in appendix A)

Folding Mechanisms

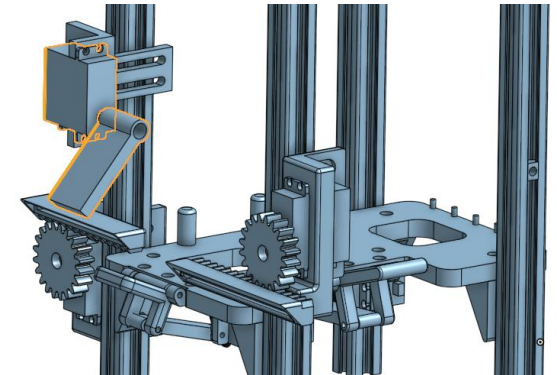
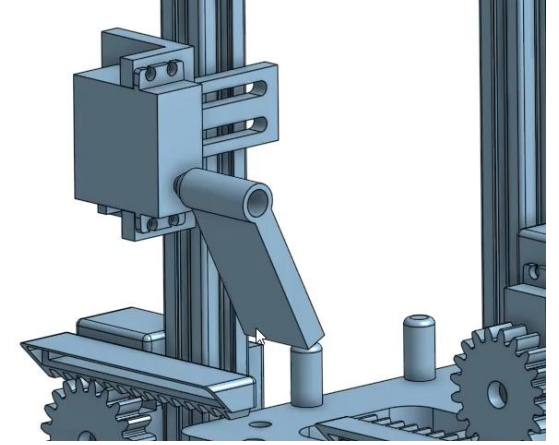
1 Linkage



2 Rack & Pinion

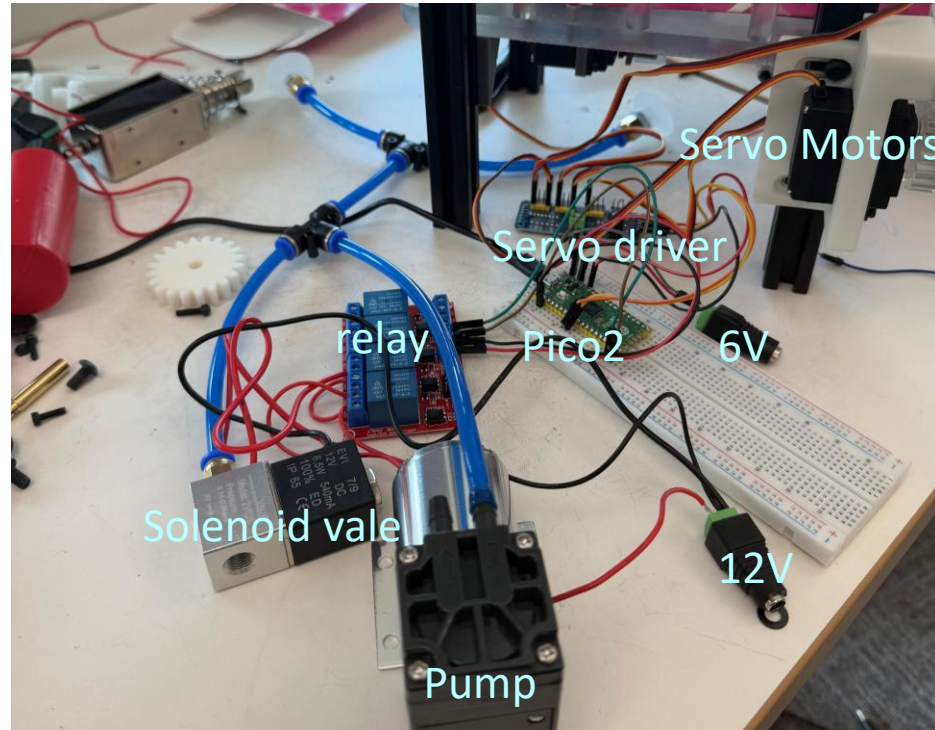
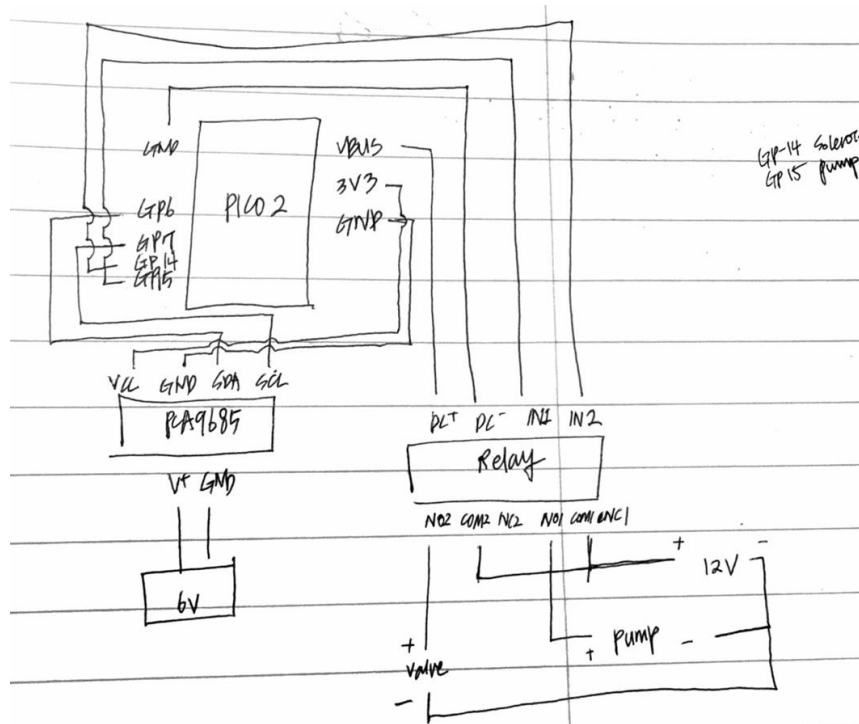


3 Rotary



Videos

Control Wiring: Design and Implementation



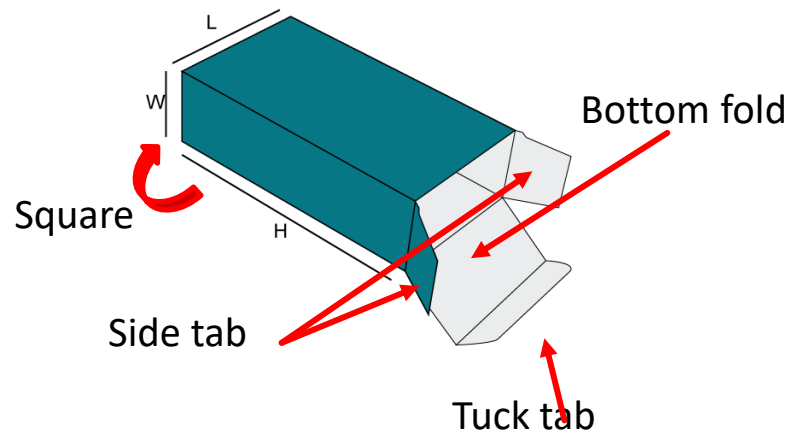
- I2C servo control – Pico2 GP6(SDA) and GP7(SCL) drives servo driver(PCA9685)
- Isolated power rail – 6V for servo motors and driver + 12V for solenoid/ pump/ valve. Common ground only
- Optocoupled relay switch for 12V load so the Pico's GPIO never carries load current directly
- Sequenced actuation than simultaneous– keep peak current within 6V/ 12V power supply

Command Reference Micropython

```
Run [stage]
  stages: square, tabs_in, fold_step1,
         tuck_in, tabs_out, fold_step2,
         tuck_out, fold_final

Set <servo> <value>
  servo: square, tab_l, tab_r, fold, tuck
  value: in degrees angle (ex) 0, 180

Pump on / off
```



Execution Sequence

- 1 Form the square
- 2 Fold side tabs in
- 3 Half of bottom fold (60degrees)
- 4 Fold tuck tab in
- 5 Pull side tab fixtures out
- 6 Second bottom fold to put the tuck tab in the box (80 degrees)
- 7 Pull the Tuck flap out
- 8 Final full bottom fold

8 total processes run in sequence for 3sec

Rapid Prototyping Progress Video



Future Work for Remainder of Internship

1 Now

Folding mechanism complete

2 Next

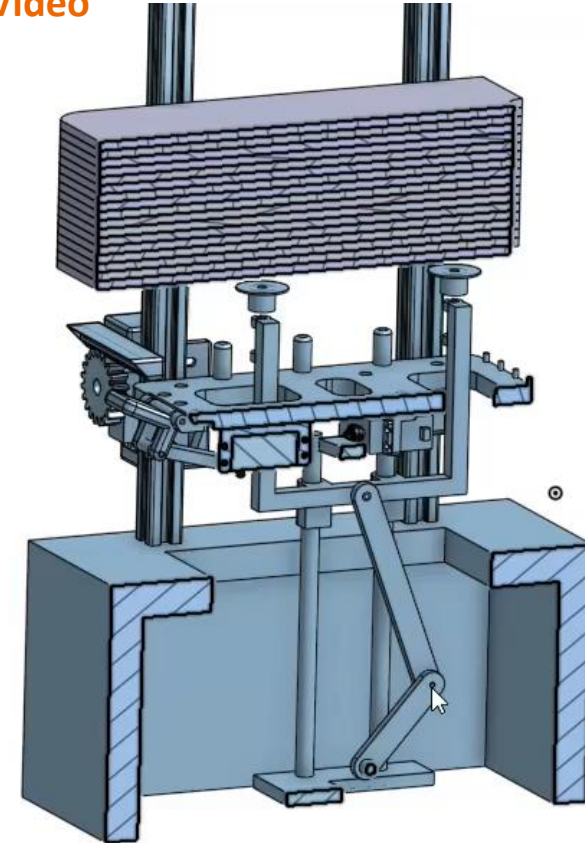
Feeding mechanism Potential Prototype - use a servo crank-slider.

3 Goal

Drop mechanism

Industrialization Move from rapid prototype to something that can be reliably used in an industry manufacturing setting.

Video



Side Projects

1 Label Applicator Back-Up Fix

Supporting different options for labeling backup process with existing semi-automated label applicator.



- Built-in through beam gap sensor misfires stop logic with current labels
- Explore options:
 - (a) redesigning label for applicator compatibility
 - (b) shifting label positioning to avoid sensor detection

2 Intelliblate IOQ Documentation

Shadowed IOQ validation process and wrote documentation.

3 Data consolidation for automated label inspection

Appendix A: BOM

Electronics				
Control				
1	Raspberry Pi Pico 2	Pre-soldered	1	\$13.43
2	PCA9685 servo driver	16 channel PWM driver, 2-pack, Compatible with Raspberry Pi	1	\$9.99
3	4-channel relay module	5V, 10A, 4 channel relay, optocoupler	1	\$9.99
Power supplies				
4	6V 3A power supply	barrel connector	1	\$8.99
5	12V 6A power supply	barrel connector	1	\$15.99
6	Barrel jack breakout	Screw terminal, Female DC power connector	1	\$6.99
Wiring & prototyping				
7	Breadboard + jumper wires	solderless breadboards + 126 pc jumpers	1	\$9.99
8	silicone wire	10ft, silicone rubber insulation	1	\$7.48
Servos				
1	MG996R servo	Metal gear, 12 kg-cm @6V, 6-pack	1	\$21.39
Solenoids				
10	Push-pull solenoid, 12V	12V, 65N, 20mm stroke, 2A, 460g	2	\$19.99
Pneumatics				
Vacuum generation				
1	Vacuum pump	12V DC, suction 5L/min, 120kPa, miniature DC motor drive,	1	\$23.77
2	Solenoid air valve	2-way Normally Closed, 1/8" NPT, 12V, zero-differential	1	\$9.99
Suction & fittings				
3	Bellows suction cup	30mm dia, M5 stud, silicone, 6-pack	1	\$7.89
4	Tube + fitting kit	6mm OD/4mm ID PU(10m) + tees, elbows, unions, PC6-M5, PC6-01/02	1	\$13.99
Frame				
1	2020 T-slot aluminum	4pcs, 400mm	1	\$18.99
Fasteners and Consumables				
1	M3nuts and screws	M3x30-50	1	\$9.99
2	nylon spacers/washers	650 pieces of flat washers, M3-12	1	\$4.74
3	M5 T-nuts and bolt kit	2020 slot, 275 pcs, t sliding nut, m5x8/10/12mm	1	\$9.99